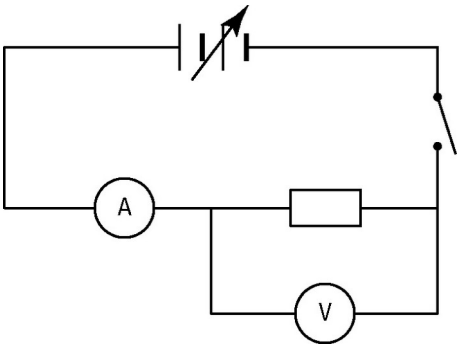
Worksheet 10.7

Science Quest 9: pages 369–71

Ohm's Law

Student: ..... Class: .....

1. The following circuit was constructed to investigate Ohm's Law using two different resistors. The results of the two experiments are shown in the table below.



Resistor	Voltage (V)	Current (A)
X	0	0
	2	2.0
	4	4.0
	6	6.0
	8	8.0
Y	0	0
	2	2.7
	4	5.3
	6	8.0
	8	10.7

(a) On the grid below plot line graphs of Current (vertical axis) versus Voltage for each resistor. Draw the line of best fit.

(b) Explain how these graphs demonstrate Ohm's Law.

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(c) Use the graphs to calculate the resistance of each resistor.

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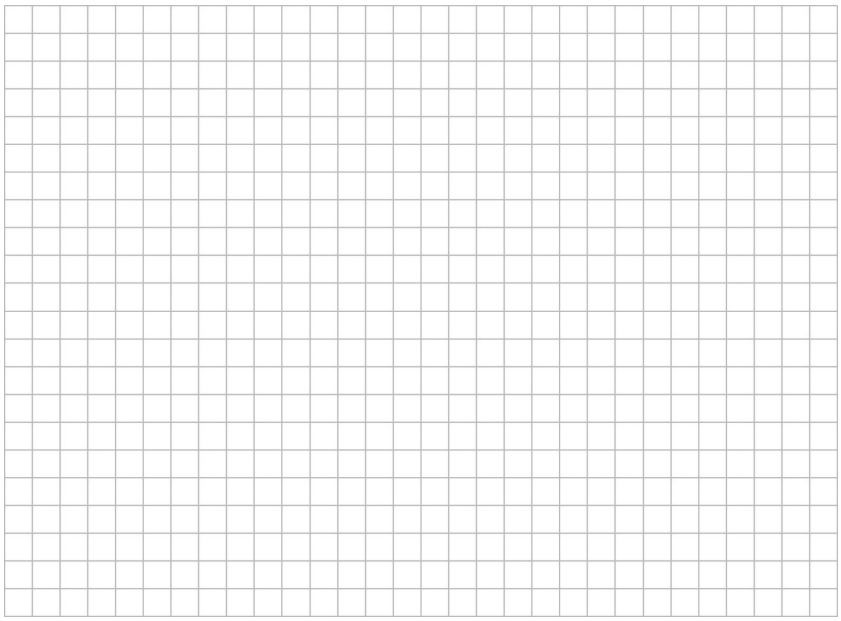
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2. The experiment was repeated but this time the resistor was replaced by a light globe. The results are shown below right.

(a) Draw a line graph of this data and draw the line of best fit.



V (V)	*I* (A)
0	0
2	2.0
4	4.0
6	5.7
8	7.0
10	7.8

(b) Describe how the current changes as the voltage increases.

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(c) Account for the trend in the data.

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